

Treatment-induced response cascade in good responders

Pre-treatment good responder:
negative · DNA-repair-proficient · $\pm$ HLA-LOH · neoantigen-positive

TNT (FOLFOX/CAPOX + CRT)

Mutation
clearance
 Δ missense -67

Neoantigen
clone elimination
 Δ binders -312

HLA-LOH
clone contraction
(S3, S4)

Immune
reprogramming
Treg+MHC-II $\uparrow$

Lymphocyte
infiltration
IGH +1,424

Pathologic complete / near-complete response (TRG 0-1)

non-responder: minimal molecular or immune change $\rightarrow$ primary treatment insensitive